# QUASI-AFFINENESS DETECTED AT A FINITE LEVEL OF THE LIMIT

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*Editorial note.* No omissions, hints or 'immediate' steps in the counted span.

*Allowed standard prerequisites (editorial).*

- Opens of the limit. Limits, Lemma lemma-descend-opens (same chapter)
- Quasi-affine criterion. Properties of Schemes, quasi-affine characterization

*Source reference.* Original reference 086P: sources/stacks-chow/tags.txt, source lines 10399–10399. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original lemma 01Z0: sources/stacks-chow/tags.txt, source lines 2575–2575. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 01PV: sources/stacks-chow/tags.txt, source lines 2244–2244. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original lemma 01Z4: sources/stacks-chow/tags.txt, source lines 2579–2579. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 01PS: sources/stacks-chow/tags.txt, source lines 2241–2241. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 01QE: sources/stacks-chow/tags.txt, source lines 2263–2263. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *In Situation 086P if $S$ is quasi-affine, then for some $i_0 \in I$ the schemes $S_i$ for $i \geq i_0$ are quasi-affine.*

*Proof.* Choose $i_0 \in I$. Note that $I$ is nonempty as the limit is directed. For convenience we write $S_0 = S_{i_0}$ and $i_0 = 0$. Let $s \in S$. We may choose an affine open $U_0 \subset S_0$ containing $f_0(s)$. Since $S$ is quasi-affine we may choose an element $a \in \Gamma(S, \mathcal{O}_S)$ such that $s \in D(a) \subset f_0^{-1}(U_0)$, and such that $D(a)$ is affine. By Lemma 01Z0 there exists an $i \geq 0$ such that $a$ comes from an element $a_i \in \Gamma(S_i, \mathcal{O}_{S_i})$. For any index $j \geq i$ we denote $a_j$ the image of $a_i$ in the global sections of the structure sheaf of $S_j$. Consider the opens $D(a_j) \subset S_j$ and $U_j = f_{j0}^{-1}(U_0)$. Note that $U_j$ is affine and $D(a_j)$ is a quasi-compact open of $S_j$, see Properties, Lemma 01PV for example. Hence we may apply Lemma 01Z4 to the opens $U_j$ and $U_j \cup D(a_j)$ to conclude that $D(a_j) \subset U_j$ for some $j \geq i$. For such an index $j$ we see that $D(a_j) \subset S_j$ is an affine open (because $D(a_j)$ is a standard affine open of the affine open $U_j$) containing the image $f_j(s)$.

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We conclude that for every $s \in S$ there exist an index $i \in I$, and a global section $a \in \Gamma(S_i, \mathcal{O}_{S_i})$ such that $D(a) \subset S_i$ is an affine open containing $f_i(s)$. Because $S$ is quasi-compact we may choose a single index $i \in I$ and global sections $a_1, \ldots, a_m \in \Gamma(S_i, \mathcal{O}_{S_i})$ such that each $D(a_j) \subset S_i$ is affine open and such that $f_i : S \to S_i$ has image contained in the union $W_i = \bigcup_{j=1,\ldots,m} D(a_j)$. For $i' \geq i$ set $W_{i'} = f_{i'i}^{-1}(W_i)$. Since $f_i^{-1}(W_i)$ is all of $S$ we see (by Lemma 01Z4 again) that for a suitable $i' \geq i$ we have $S_{i'} = W_{i'}$. Thus we may replace $i$ by $i'$ and assume that $S_i = \bigcup_{j=1,\ldots,m} D(a_j)$. This implies that $\mathcal{O}_{S_i}$ is an ample invertible sheaf on $S_i$ (see Properties, Definition 01PS) and hence that $S_i$ is quasi-affine, see Properties, Lemma 01QE. Hence we win. $\square$